RILSAMID®

TECHNICAL DATA SHEET

# RILSAMID® ASR 13

## POLYAMIDE 12 PELLET

RILSAMID® ASR 13 a polyamide 12 compound. This black reinforced grade is designed for injection molding & is used to produce automotive fittings & quick connectors, the incorporation of carbon fiber leading to a product with same rigidity as a 30% glass fiber reinforced grade & electrostatic discharge (ESD) performance for conductive parts.

**Designation :** ISO 16396 - PA12, CF15, M1G1HL, C16-080

### TYPE

PA12-CF15

### POTENTIAL APPLICATIONS

- Chemical & plastic distribution
- Chemicals manufacturing
- Commercial truck
- Extrusion 3d
- Light vehicule

### DELIVERY FORM

- Pellets

### TRANSFORMATION PROCESSES

- Injection Molding

### ADDITIVES

- Heat Stabilized
- Light Stabilized
- Release agent

## RHEOLOGICAL PROPERTIES

| PROPERTIES | TYPICAL VALUE | UNIT | TEST STANDARD |
|---|---|---|---|
| Melt volume flow rate (MVR), 235°C / 5 kg (455°F / 11 lb) | 10 | $cm^3$/10min | ISO 1133 |
| Shrinkage, parallel (t+24h) | 0.2 | % | ISO 294-4 |
| Shrinkage, normal (t+24h) | 0.5 | % | ISO 294-4 |

## MECHANICAL PROPERTIES

| PROPERTIES | TYPICAL VALUE | UNIT | TEST STANDARD |
|---|---|---|---|
| Flexural modulus, 23°C (73°F) | - / 7200 | MPa | ISO 178 |
| Tensile modulus, 23°C (73°F) | 9600 / 9000 | MPa | ISO 527-1/-2 |
| Stress at break, 23°C (73°F) | - / 110 | MPa | ISO 527-1/-2 |
| Strain at break, 23°C (73°F) | - / 4 | % | ISO 527-1/-2 |
| Charpy unnotched impact strength, 23°C (73°F) | 45 / 46 | kJ/m2 | ISO 179 1eU |
| Charpy unnotched impact strength, -30°C (-22°F) | 47 / 48 | kJ/m2 | ISO 179 1eU |
| Charpy notched impact strength, 23°C (73°F) | - / 11 | kJ/m2 | ISO 179 1eA |
| Charpy notched impact strength, -30°C (-22°F) | 8 / 7 | kJ/m2 | ISO 179 1eA |

* DRY / COND
DRY: Dry As Molded (DAM) if pellet / Dry if powder.
COND: Conditionned.

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## THERMAL PROPERTIES

| PROPERTIES | TYPICAL VALUE | UNIT | TEST STANDARD |
|---|---|---|---|
| Melting temperature, 10°C/min | 178 | °C | ISO 11357-1/-3 |
| Thermal degradation temperature, Under air at 10°C/min (1%_wt_Loss) | 223 | °C | ISO 11358-1 |
| Thermal degradation temperature, Under nitrogen at 10°C/min (1%_wt_Loss) | 223 | °C | ISO 11358-1 |

## ELECTRICAL PROPERTIES

| PROPERTIES | TYPICAL VALUE* | UNIT | TEST STANDARD |
|---|---|---|---|
| Surface resistivity, 23°C (73,4°F) | - / 1.0E+05 | ohm/sq | IEC 62631-3-2 |

* DRY / COND
DRY: Dry As Molded (DAM) if pellet / Dry if powder.
COND: Conditionned.

## OTHER PROPERTIES

| PROPERTIES | TYPICAL VALUE | UNIT | TEST STANDARD |
|---|---|---|---|
| Water absorption, at equilibrium at 23°C (73°F) / 50%RH | 0.6 | % | ISO 62 |
| Density, 23°C (73°F) | 1.08 | g/cm³ | ISO 1183-1 |
| Water absorption, at equilibrium at 23°C (73°F) / Immersion | 1.5 | % | ISO 62 |

## SHELF LIFE

Two years from the date of delivery, when stored properly (sealed bags, appropriate moisture, UV protection and temperature). For any use above this limit, please refer to our technical services.

## PROCESSING RECOMMENDATIONS

Please refer to specific grade's processing guidelines

- Typical melt temperature (Min / Recommended / Max) - Injection Molding: 240°C / 260°C / 280°C (465°F / 500°F / 535°F)
- Typical mold temperature - Injection molding: 40-90°C (105-195°F)
- Drying Conditions (only for bags opened for more than 2 hours): 80-90°C (175-195°F) / 4-6 hours

## SPECIAL CHARACTERISTICS

- Electrostatic discharge (ESD)

## REGIONAL AVAILABILITY

World



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